

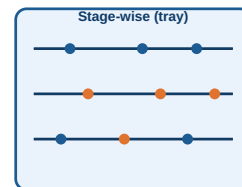
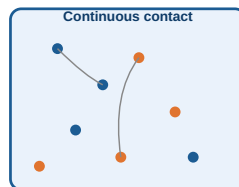
CHE213 — Mass Transfer Roadmap

Chemical Engineering Department | Textbooks: Treybal • Geankoplis • McCabe/Smith/Harriott | NPTEL • MIT OCW • LearnChemE

1 FUNDAMENTALS OF MASS TRANSFER & MOLECULAR DIFFUSION

1.1 Mass Transfer Approaches

- **Rate-Based Approach** – models continuous-contact equipment (packed columns, film devices) using local fluxes and transfer coefficients, without assuming interphase equilibrium.
- **Equilibrium-Based Approach** – assumes the streams leaving each stage reach complete thermodynamic equilibrium; used for discrete stage-wise equipment (trayed towers).



Rate-based (continuous contact) vs. Equilibrium-based (stage-wise) equipment

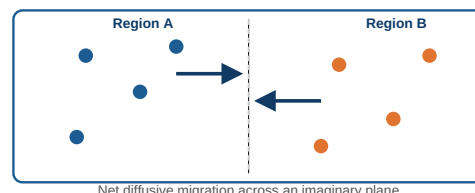
1.2 Nomenclature & Kinematics

ρ_i – mass concentration of i (kg/m^3) | C_i – molar concentration of i (kmol/m^3)

$\omega_i = \rho_i/\rho$ – mass fraction | $x_i = C_i/C$ (liquid), $y_i = P_i/P$ (ideal gas) – mole fraction

Mass-avg velocity: $u = \sum \omega_i u_i$ **Molar-avg velocity:** $U = \sum x_i u_i$

Absolute flux: $N_i = C_i u_i$ (fixed frame) **Relative diffusive flux:** $J_i = C_i(u_i - U)$ (moving w/ U)



Net diffusive migration across an imaginary plane

Species velocities & fluxes relative to a fixed plane

1.3 Fick's First Law of Diffusion & Flux Equations

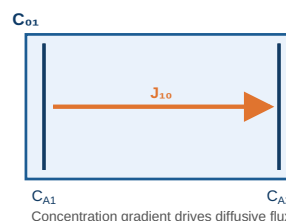
$$J_A = -D_{AB} \frac{dC_A}{dz} \quad (1)$$

Diffusive flux J_A is proportional to the concentration gradient dC_A/dz ; D_{AB} is the diffusivity of A in B (m^2/s).

- **Gas-phase diffusivity** (Chapman–Enskog): $D_{AB} \propto T^{3/2}/(P\sqrt{M_A M_B} \cdot \sigma_{AB}^2 \Omega_D)$ – strong $T^{3/2-1.75}$ dependence.
- **Liquid-phase diffusivity** (Wilke–Chang): $D_{AB}^{\circ} = 7.4 \times 10^{-8} (\phi M_B)^{1/2} T / (\mu_B V_A^{0.6})$ – scales inversely with solvent viscosity μ_B .

$$N_A = x_A(N_A + N_B) - D_{AB} \frac{dC_A}{dz} \quad (2)$$

Total flux = Convective contribution $x_A(N_A + N_B)$ + Molecular diffusive contribution $-D_{AB}dC_A/dz$.



Concentration gradient drives diffusive flux

Fick's Law – linear concentration profile across a stagnant film

Phase 1 Learning Resources

NPTEL: Mass Transfer Operations I – Molecular Diffusion in Gases and Liquids • **LearnChemE:** Fundamentals of Diffusion Screencasts

2 SPECIES BALANCE & SPECIAL DIFFUSION CASES

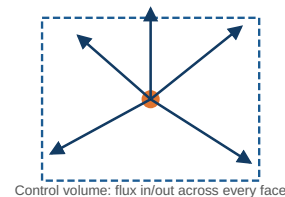
2.1 Species Continuity Equation & Fick's Second Law

$$\partial C_A / \partial t + \nabla \cdot N_A - R_A = 0 \quad (3)$$

Cartesian form: $\partial C_A / \partial t + (\partial N_{A,x} / \partial x + \partial N_{A,y} / \partial y + \partial N_{A,z} / \partial z) - R_A = 0$ — accumulation, net outward flux, and reaction generation/consumption.

$$\partial C_A / \partial t = D_{AB} \nabla^2 C_A \quad (4)$$

Fick's 2nd Law – stationary medium ($U = 0$), no reaction ($R_A = 0$).



Control volume: flux in/out across every face

Differential control volume for the species balance

2.2 1-D Steady-State Diffusion Case Studies

Equimolar Counterdiffusion (EMCD): $N_A = -N_B$, so $N_A + N_B = 0$ (no net bulk convective velocity).

$$N_A = \frac{D_{AB}}{RTz} (P_{A,1} - P_{A,2}) \quad (5)$$

Partial-pressure profile is linear across diffusion length z .

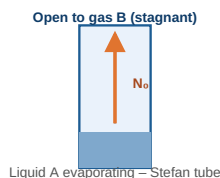
Diffusion of A through Non-diffusing / Stagnant B: $N_B = 0$ (B insoluble/blocked), inducing forward bulk flow.

$$N_A = \frac{D_{AB}P}{RTzP_{BM}} (P_{A,1} - P_{A,2}) \quad (6)$$

$$P_{BM} = \frac{P_{B,2} - P_{B,1}}{\ln(P_{B,2}/P_{B,1})} \quad (7)$$



Equimolar counterdiffusion between two bulbs

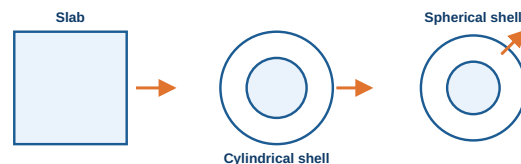


Diffusion of A through stagnant film of B

2.3 Diffusion in Differing Geometries

Geometry	Area A	Molar Rate W_A	Flux Profile
Flat Slab	Constant (A_z)	Constant	N_A constant along z
Cylindrical Shell	$A_r = 2\pi rL$	Constant	$N_{A,r} \propto 1/r$
Spherical Shell	$A_r = 4\pi r^2$	Constant	$N_{A,r} \propto 1/r^2$

Molar rate $W_A = N_A A$ stays constant along the diffusion path at steady state (mass conservation); only the local flux N_A changes with area.



Diffusion geometries: slab, cylindrical shell, spherical shell

Phase 2 Learning Resources

NPTEL: Mass Transfer Operations I – Steady-State Diffusion Case Studies • **MIT OCW:** Mass Transfer Continuity Equations

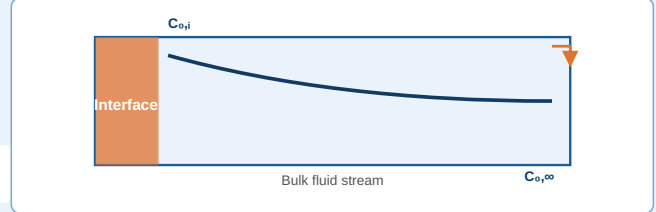
3 CONVECTIVE MASS TRANSFER & MASS TRANSFER COEFFICIENTS

3.1 Convective Mechanisms & MTC Definition

- **Natural Convection** – fluid motion driven by density gradients from concentration/temperature differences.
- **Forced Convection** – fluid motion driven by external forces (pumps, impellers, pressure drop).

$$N_A = k_c(C_{A,i} - C_{A,\infty}) \quad (8)$$

Mass transfer coefficient k_c relates flux to the overall driving force across the boundary layer.



Interphase mass transfer – concentration profile from interface to bulk

3.2 MTC Definitions & Equivalences

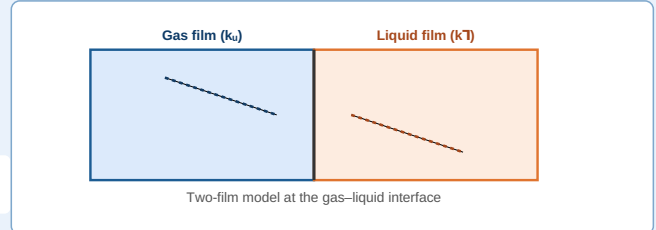
Phase	Driving Force	Symbol	Flux Equation	Conversion
Gas	ΔP_A	k_G	$N_A = k_G(P_{A,1} - P_{A,2})$	$k_c = k_G RT$
Gas	Δy_A	k_y	$N_A = k_y(y_{A,1} - y_{A,2})$	$k_y = k_G P = k_c C$
Gas/Liquid	ΔC_A	k_c, k_L	$N_A = k_c(C_{A,1} - C_{A,2})$	$k_c = D_{AB}/\delta$
Liquid	Δx_A	k_x	$N_A = k_x(x_{A,1} - x_{A,2})$	$k_x = k_L C$

3.3 Boundary Layer Theories & Overall MTC

- **Film Theory** – a stagnant film of thickness δ where transport is purely diffusive: $k_c = D_{AB}/\delta$.
- **Penetration / Surface-Renewal Theory** – transient diffusion into fluid elements continuously replaced at the interface: $k_c \propto \sqrt{D_{AB}}$.

$$1/K_G = 1/k_G + m/k_L \quad (9)$$

Overall MTC K_G combines gas- and liquid-film resistances via the Henry's-Law slope m – the two-film resistance analogy.

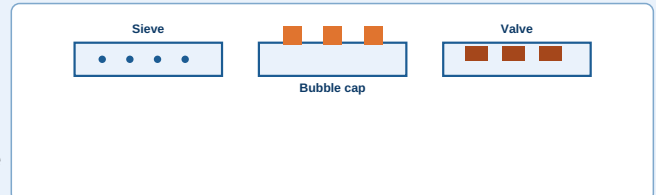


Two-film theory: series resistance across the interface

4 CONTACTING EQUIPMENT – STAGE-WISE COLUMNS

4.1 Tray (Stage-Wise) Columns

- **Sieve Trays** – flat perforated plates; simple, low cost, lowest ΔP , limited turndown (weeping).
- **Bubble Cap Trays** – standpipes with capped risers; stable at low flow, higher cost/ ΔP .
- **Valve Trays** – variable-orifice units; balance low ΔP , moderate cost, wide turndown.



Sieve, bubble-cap, and valve tray decks

Weeping

Liquid drains through holes – vapor velocity too low.

Flooding

Excess liquid buildup – gas/liquid rates too high.

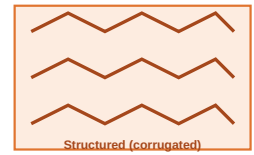
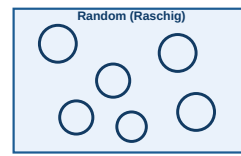
Entrainment

Liquid carried upward by excessive vapor velocity.

4 CONTACTING EQUIPMENT – CONTINUOUS CONTACT COLUMNS (cont.)

4.2 Packed Beds

- **Random Packings** – dumped elements (Raschig rings, Pall rings, Intalox saddles); high area, high voidage.
- **Structured Packings** – corrugated wire-mesh/sheet-metal matrices; optimize area per unit volume.
- **Design Parameters** – interfacial area (a), packing factor (F_p) for GPDC correlations, liquid holdup, redistribution every 3–5 bed diameters.



Feature	Tray (Stage-wise) Column	Packed (Continuous-contact) Column
Contact mode	Discrete equilibrium stages	Continuous differential contact
Pressure drop	Higher, per tray	Generally lower
Best suited for	High liquid loads, large diameters	Corrosive service, low ΔP needs, small diameters
Design basis	Number of trays (N)	Height $Z = HTU \times NTU$

5 GAS ABSORPTION & STRIPPING OPERATIONS

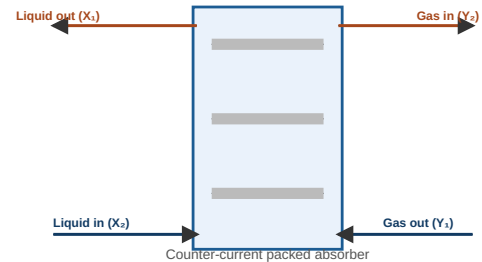
5.1 Operating Lines & Packed Tower Height

$$L_s(X_2 - X_1) = G_s(Y_2 - Y_1) \quad (10)$$

Solute-free mole-ratio balance: $X = x/(1-x)$, $Y = y/(1-y)$; L_s , G_s = inert molar flow rates.

$$Z = HTU \times NTU = H_{OG} \times N_{OG} \quad (11)$$

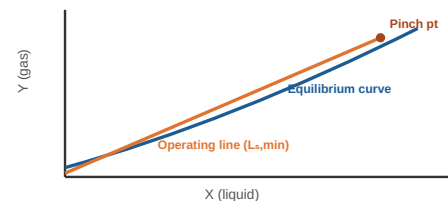
$$H_{OG} = \frac{G_s}{K_{OG}aP}, \quad N_{OG} = \int_{Y_2}^{Y_1} \frac{dY}{Y - Y^*} \quad (12)$$



Packed absorption column – counter-current gas/liquid contact

5.2 Tray Absorber Design Constraints

- **Minimum Solvent Flow ($L_{s,min}$)** – where the operating line touches the equilibrium curve ($Y^* = f(X)$) at an infinite-stage pinch point.
- **Design Solvent Rate** – typically $1.25\text{--}1.5 \times L_{s,min}$ to balance capital vs. operating cost.



Minimum liquid rate at the operating-line / equilibrium pinch

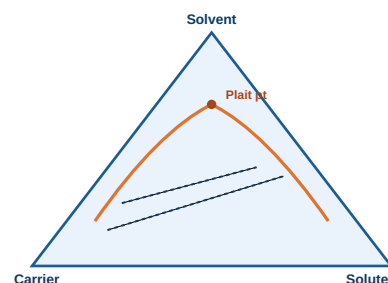
Phase 3–5 Learning Resources

NPTEL: Convective MTCs & Interphase Transport • Equipment Design & Hydraulics • Absorption Column Design • **LearnChemE:** MTC, Column Hydraulics & Packed Tower NTU/HTU Screencasts

6 LIQUID-LIQUID EXTRACTION (LLX) & DISTILLATION

6.1 Liquid-Liquid Extraction

- **Ternary Equilibrium Coordinates** – triangular diagrams mapping solute/carrier/solvent compositions, binodal solubility curves, plait points, and tie lines.
- **Multi-Stage Configurations** – single-stage, cross-current contactors, and continuous countercurrent cascades.



Ternary diagram: binodal curve, tie lines, and plait point

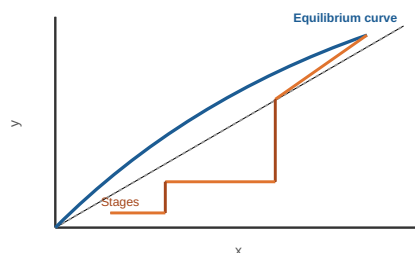
6.2 Distillation & McCabe-Thiele Method

$$\alpha_{AB} = \frac{y_A/x_A}{y_B/x_B} = \frac{P^\circ_A y_A}{P^\circ_B y_B} \quad (13)$$

$$y_{n+1} = \frac{R}{R+1} x_n + \frac{x_D}{R+1} \quad (\text{ROL}, R = L/D) \quad (14)$$

$$y_{m+1} = \frac{L_m}{V_m} x_m - \frac{W x_W}{V_m} \quad (\text{SOL}) \quad (15)$$

$$y = \frac{q}{q-1} x - \frac{x_F}{q-1} \quad (\text{q-line}) \quad (16)$$



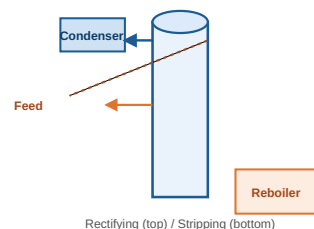
McCabe-Thiele stage stepping between x_W and x_D

Feed Condition	q Value	q-Line Slope
Cold subcooled liquid	$q > 1$	Positive (> 1)
Saturated liquid (bubble pt.)	$q = 1$	Vertical (∞)
Partially vaporized	$0 < q < 1$	Negative
Saturated vapor (dew pt.)	$q = 0$	Horizontal (0)
Superheated vapor	$q < 0$	Positive (< 1)

6.3 Limiting Reflux Conditions

- **Minimum Reflux ($R_{\min}$)** – ROL, SOL, q-line coincide on the VLE curve at an infinite-stage pinch point.
- **Total Reflux ($R = \infty$)** – gives the minimum theoretical stages via the Fenske equation.

$$N_{\min} = \frac{\ln[(x_D/(1-x_D))((1-x_W)/x_W)]}{\ln(\alpha_{\text{avg}})} \quad (17)$$



Distillation column: condenser, reflux, feed, and reboiler

Phase 6 Learning Resources

NPTEL: Mass Transfer Operations II – Distillation Methods • LearnChemE: Interactive McCabe-Thiele Simulations

CHE213 — Practice, Revision & PYQ Resources

CORE TEXTBOOKS & DIGITAL RESOURCES

- Treybal, R. E. – *Mass-Transfer Operations*, 3rd Ed., McGraw-Hill.
- Geankoplis, C. J. – *Transport Processes and Separation Process Principles*, 4th Ed., Prentice Hall.
- McCabe, W. L., Smith, J. C., & Harriott, P. – *Unit Operations of Chemical Engineering*, 7th Ed., McGraw-Hill.
- NPTEL Course: Mass Transfer Operations I (IIT Guwahati)
- NPTEL Course: Mass Transfer Operations II (IIT Guwahati)
- MIT OpenCourseWare: Transport Processes (10.302)
- LearnChemE: Mass Transfer & Separations Screencasts

ROADMAP AT A GLANCE

Phase	Topic	Key Tools
1	Fundamentals & Molecular Diffusion	Fick's 1st Law, Chapman–Enskog, Wilke–Chang
2	Species Balance & Diffusion Cases	Fick's 2nd Law, EMCD, Stagnant-film diffusion
3	Convective Mass Transfer & MTCs	Film theory, Penetration theory, Two-film model
4	Contacting Equipment	Tray types, Random/Structured packing
5	Gas Absorption & Stripping	HTU–NTU, Operating line, $L_{S,min}$
6	LLX & Distillation	Ternary diagrams, McCabe–Thiele, Fenske equation

SUPPLEMENTARY: KEY DIMENSIONLESS GROUPS

Group	Definition	Physical Meaning
Reynolds, Re	$Re = \rho u L / \mu$	Inertial vs. viscous forces (flow regime)
Schmidt, Sc	$Sc = \mu / (\rho D_{AB})$	Momentum vs. mass diffusivity
Sherwood, Sh	$Sh = k_c L / D_{AB}$	Convective vs. diffusive mass transfer
Lewis, Le	$Le = Sc / Pr = \alpha / D_{AB}$	Thermal vs. mass diffusivity

$Sh = f(Re, Sc)$ e.g. $Sh = 0.023 Re^{0.8} Sc^{1/3}$ (turbulent pipe flow) (18)

Chilton–Colburn analogy links friction, heat and mass transfer: $j_D = Sh / (Re \cdot Sc^{1/3})$
 $= j_H = f/2$, allowing k_c to be estimated from known heat-transfer or friction data.



Links friction, heat & mass transfer coefficients

Transfer-analogy triangle: momentum ↔ heat ↔ mass

CONCEPT FLOW OF THE ROADMAP



Previous Year Questions (PYQs)

Use these for practice and revision alongside the roadmap above:

- CHE213 PYQ Drive Folder: https://drive.google.com/drive/folders/1D3G6EWzg_kKxhGOJho1eKOxWVVHwqmqKO?usp=share_link